



Exercise Session 10 Highlights

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Course: Reglerteknik (EL1000)

2023

Solution of the State Space Equations

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t), \quad x(0) = x_0 \\ y(t) &= Cx(t)\end{aligned}$$

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau) d\tau$$

$$e^{At} = I + At + \frac{A^2t^2}{2!} + ..$$

→ usually used for proofs....

$$e^{At} = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}$$

✓

$$A = TDT^{-1}, \quad D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$
$$e^{At} = Te^{Dt}T^{-1}$$
$$e^{Dt} = \begin{pmatrix} e^{\lambda_1 t} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_n t} \end{pmatrix}$$

when A is diagonalizable

Controllability

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \quad (6.1)$$

Handwritten annotations: $\mathbf{A}$ is $n \times n$, $\mathbf{B}$ is $n \times m$, $\mathbf{x}$ is $n \times 1$, and $\mathbf{u}$ is $m \times 1$.

Definition 6.1 The state equation (6.1) or the pair $(\mathbf{A}, \mathbf{B})$ is said to be controllable if for any initial state $\mathbf{x}(0) = \mathbf{x}_0$ and any final state $\mathbf{x}_1$, there exists an input that transfers $\mathbf{x}_0$ to $\mathbf{x}_1$ in a finite time. Otherwise (6.1) or $(\mathbf{A}, \mathbf{B})$ is said to be uncontrollable.

This definition requires only that the input be capable of moving any state in the state space to any other state in a finite time; what trajectory the state should take is not specified. Furthermore, there is no constraint imposed on the input; its magnitude can be as large as desired.

Checking controllability:

$$\mathcal{S} = [B, AB, \dots, A^{n-1}B] \quad n \times (nm)$$

$\mathcal{S}$ should have full row rank $\rightarrow \text{Rank}(\mathcal{S}) = n$

Controllable state vector: A state vector is said to be controllable if there is an input signal that brings the state $\mathbf{x}$ from the origin to $\mathbf{x}^*$ in finite time.

The controllable states can be written as linear combinations of matrix $\mathcal{S}$ columns (i.e., they are spanned by columns of $\mathcal{S}$)

Observability

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \\ \mathbf{y} &= \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}\end{aligned}\tag{6.22}$$

Handwritten dimension annotations: $\mathbf{A}$ is $n \times n$, $\mathbf{B}$ is $n \times m$, $\mathbf{C}$ is $p \times n$, and $\mathbf{D}$ is $p \times m$.

Definition 6.01 The state equation (6.22) is said to be observable if for any unknown initial state $\mathbf{x}(0)$, there exists a finite $t_1 > 0$ such that the knowledge of the input $\mathbf{u}$ and the output $\mathbf{y}$ over $[0, t_1]$ suffices to determine uniquely the initial state $\mathbf{x}(0)$. Otherwise, the equation is said to be unobservable.

Checking Observability:

O should have full column rank

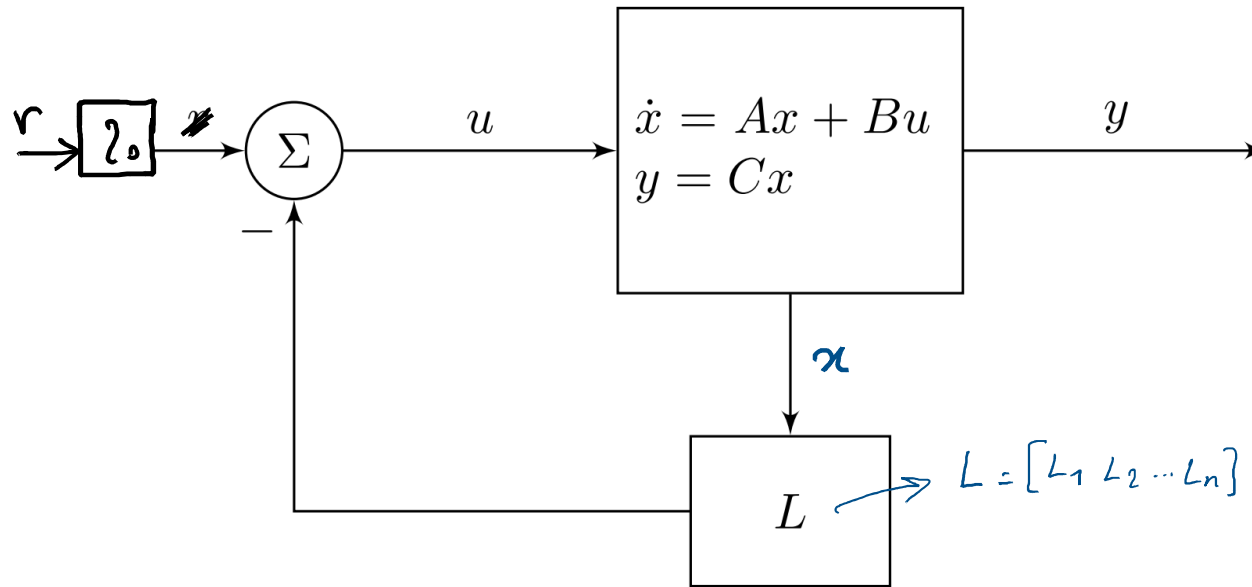
$$\hookrightarrow \text{Rank}(\mathcal{O}) = n$$

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} \quad (np) \times n$$

Unobservable state vector: A state vector $\mathbf{x}^* \neq 0$ is said to be unobservable if the output is identically zero when $\mathbf{x}(0) = \mathbf{x}^*$ and $\mathbf{u}(t) = 0$.

The unobservable state vectors are in the null space of matrix O

State Feedback



if (A, B) is controllable we can choose L in the state feedback so that the closed-loop system gets arbitrary placed poles.

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \\ u = -Lx + l_0 r \end{cases} \Rightarrow \begin{cases} \dot{x} = Ax + B(-Lx + l_0 r) \\ y = Cx \end{cases}$$

State feedback $\hookrightarrow$

$$\Rightarrow \begin{cases} \dot{x} = (A - BL)x + Bl_0 r \\ y = Cx \end{cases}$$

closed loop system.

Poles:

$$\det(sI - A + BL) = 0$$

State Feedback (cont.)

For tracking a constant reference signal $\bar{r}$ in the output we should tune l_0 properly so that the closed-loop transfer function has a DC gain equal to 1 (when there is no external disturbance):

$$G_c(s) = C(sI - (A - BL))^{-1} B l_0$$

$$\begin{cases} \dot{x} = (A - BL)x + B l_0 r \\ y = Cx \end{cases} \Rightarrow G_c(s) = \frac{(b_1 s^{n-1} + \dots + b_n) l_0}{s^n + \alpha_1 s^{n-1} + \dots + \alpha_n}$$

$$G_c(0) = 1 \Rightarrow l_0 = \frac{\alpha_n}{b_n}$$

$$\lim_{s \rightarrow 0} s G_c(s) \frac{\bar{r}}{s} = \bar{r}$$

when $G_c(s)$ does not have a zero at the origin.

* Alternatively: Consider the closed loop system at the steady state, i.e., Let $\dot{x} = 0$

Sometime this is easier!

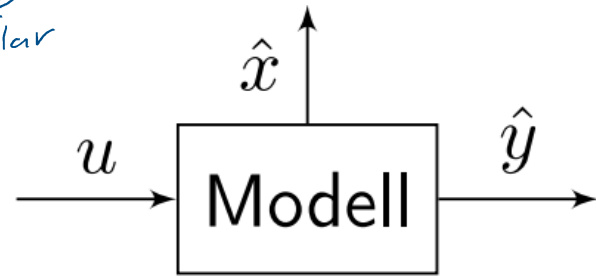
and find l_0 such that $y = \bar{r}$

Observer Design

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

Scalar

⑦ when (A, C) is observable we can choose K so that the matrix $(A - KC)$ gets arbitrary placed eigenvalues.



$$\Leftrightarrow \begin{cases} \dot{\hat{x}} = A\hat{x} + Bu \\ \hat{y} = C\hat{x} \end{cases}$$

$$K = \begin{bmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{bmatrix}$$

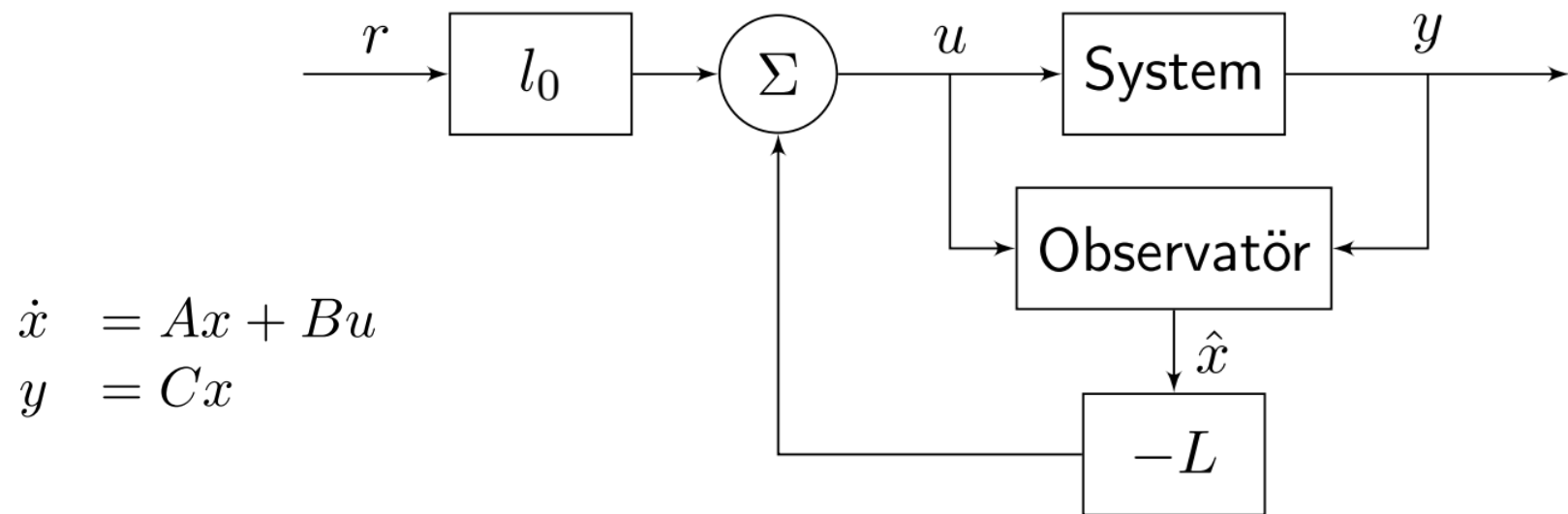
$$\dot{\hat{x}} = A\hat{x} + Bu + \overbrace{K(y - \underbrace{C\hat{x}}_{\hat{y}})}^{\text{correction term}} \rightarrow \text{Observer dynamic} \quad (\text{usually } x(0) \neq \hat{x}(0))$$

estimation error

$$\tilde{\dot{x}} = \dot{x} - \dot{\hat{x}} = Ax + Bu - A\hat{x} - Bu - K(Cx - C\hat{x})$$

$$\Rightarrow \tilde{\dot{x}} = (A - KC)\tilde{x}, \quad \text{estimation error dynamics}$$

State Feedback using an Observer!



$$\dot{\hat{x}} = A\hat{x} + Bu + K(y - c\hat{x})$$

$$u = -L\hat{x} + l_0r$$

Observer dynamic should be
faster than the closed loop :

$$|\text{eig}(A - BL)| \leq |\text{eig}(A - KC)|$$

response ! $\Rightarrow$ observer poles should be further to the left
in the L.H.P.

References

- Lectures of Automatic Control, General Course (EL1000), KTH, 2021
- Rebecka Winqvist's notes for Automatic Control, General Course (EL1000), KTH, 2021-2022
- Chen CT. Linear System Theory and Design, Oxford University Press, 1999

Useful Videos on Full State Feedback, etc. (Optional):



You can also watch **Brian Douglas'** (newer) videos:
(these videos might help you to better grasp the concepts but are not designed for solving exercises)



Full State Feedback



Controllability & Observability